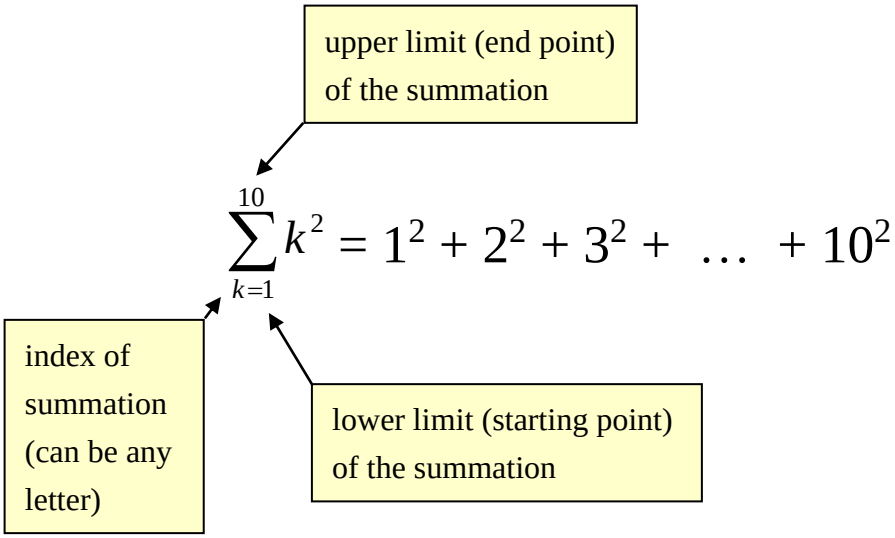
Summation notation

Summation notation (or Σ notation) is a concise and convenient way to represent long sums. For example:



The symbol Σ is the Greek capital letter Sigma corresponding to our letter 'S' and refers to the initial letter of the word 'Sum'. Therefore, this expression means the sum of all the numbers with ***general term*** k^2, where k takes the values from 1 to 10.

Note:

Besides Sigma "Σ", we can use another Greek capital letter Pi "Π" to represent long product. For example:

$$\prod_{k=1}^{10} k^2 = 1^2 \times 2^2 \times 3^2 \times \ \ldots \ \times 10^2$$

The discussion of the product notation is not in the syllabus of Module 2 but students are allowed to use it if necessary. For more information, please refer to the Assessment Framework for HKDSE Mathematics in HKEAA website.

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Summation with Σ notation

Here are some examples to use the Σ notation to represent the summation of sequences:

	Sequence	General term	No. of terms	Summation with Σ notation
e.g.1	$2 + 4 + 6 + 8 + 10$	$2k$	5	$\sum_{k=1}^{5} 2k$
e.g.2	$1 + 3 + 5 + 7 + 9 + 11$	$2k - 1$	6	$\sum_{k=1}^{6} (2k-1)$
e.g.3	$\frac{2}{3}+\frac{4}{5}+\frac{6}{7}...+\frac{50}{51}$	$\frac{2k}{2k+1}$	50	$\sum_{k=1}^{50} \frac{2k}{2k+1}$
e.g.4	$2 + 4 + 8 + 16 + 32$	2^k	5	$\sum_{k=1}^{5} 2^k$
e.g.5	$-2 + 4 - 8 + 16 - 32$	$(-1)^k 2^k$	5	$\sum_{k=1}^{5} (-1)^k 2^k$
e.g.6	$2 - 4 + 8 - 16 + 32$	$(-1)^{k+1} 2^k$	5	$\sum_{k=1}^{5} (-1)^{k+1} 2^k$
e.g.7	$1+\frac{1}{2}+\frac{1}{3}+...+\frac{1}{10}$	$\frac{1}{k}$	10	$\sum_{k=1}^{10} \frac{1}{k}$

Note that we can express summation of a sequence with different representations. For example, the following notation represent the same summation $1+\frac{1}{2}+\frac{1}{3}+...+\frac{1}{10}$:

$$\sum_{k=1}^{10} \frac{1}{k} \quad = \quad 1+\sum_{k=2}^{10} \frac{1}{k} \quad = \quad 1+\sum_{k=1}^{9} \frac{1}{k+1}$$

Of course, we will usually choose to represent the summation of sequences using the simplest form of Σ notation.

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Summation with Σ notation

Try to use the Σ notation to represent the summation of sequences:

	Sequence	General term	No. of terms	Summation with Σ notation
Q.1	$3 + 6 + 9 + 12 + 15 + 18$			
Q.2	$2 + 5 + 8 + 11 + 14 + 17$			
Q.3	$6 + 10 + 14 + 18 + 22$			
Q.4	$3 + 7 + 11 + \ldots + 75$			
Q.5	$\frac{1}{3\times5}+\frac{1}{5\times7}+\frac{1}{7\times9}\ldots+\frac{1}{99\times101}$			
Q.6	$-1^2 + 2^2 - 3^2 + 4^2 - \ldots + 20^2$			
Q.7	$\frac{1}{3}-\frac{2}{4}+\frac{3}{5}-\ldots+\frac{98}{100}$			
Q.8	$1 + 1 + 2 + 3 + 5 + 8 + \ldots + 55$			

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Summation with Σ notation

It is easier to use the Σ notation to represent the summation of sequences with given general term (usually the last term of the sequence). Here are some examples:

	Sequence	No. of terms	Summation with Σ notation
e.g.1	$1 + 4 + 9 + \ldots + n^2$	n	$\sum_{k=1}^{n} k^2$
e.g.2	$2 + 4 + 6 + \ldots + 2n$	n	$\sum_{k=1}^{n} 2k$
e.g.3	$1 + 3 + 5 + \ldots + (2n - 1)$	n	$\sum_{k=1}^{n} (2k-1)$
e.g.4	$1 - 4 + 9 - \ldots + (-1)^n n^2$	n	$\sum_{k=1}^{n} (-1)^k k^2$
e.g.5	$1-\frac{1}{2}+\frac{1}{4}-\frac{1}{8}+\ldots+(-1)^n\frac{1}{2^n}$	n	$\sum_{k=1}^{n} (-1)^k \frac{1}{2^k}$

Sometimes, we may use different Σ notation to represent the summation of same sequence. For example:

	Sequence	No. of terms	Summation with Σ notation
e.g.6	$1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\ldots+\frac{1}{2n-1}-\frac{1}{2n}$	$2n$	$\sum_{k=1}^{2n} (-1)^k \frac{1}{2k}$
		n	$\sum_{k=1}^{n} \left(\frac{1}{2k-1}-\frac{1}{2k}\right)$

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Summation with Σ notation

Try to use the Σ notation to represent the summation of the following sequence.

	Sequence	No. of terms	Summation with Σ notation
Q.1	$1 + 8 + 27 + \ldots + n^3$		
Q.2	$3 + 6 + 9 + \ldots + 3n$		
Q.3	$\frac{1}{3}+\frac{2}{3^2}+\frac{3}{3^3}+\ldots+\frac{n}{3^n}$		
Q.4	$2 + 4 + 8 + \ldots + 2^n$		
Q.5	$1+\frac{1}{1\times 4}+\frac{1}{4\times 7}+\ldots+\frac{1}{(3n-2)(3n+1)}$		
Q.6	$1+(x+1)+(x+1)^2+\ldots+(x+1)^n$		
Q.7	$\frac{1}{x-1}-\frac{1}{x}-\frac{1}{x^2}-\ldots-\frac{1}{x^n}$		
Q.8	$1^2 - 2^2 + 3^2 - 4^2 \ldots + (2n-1)^2 - (2n)^2$		

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Summation with Σ notation

Conversely, we can write out what is meant by summation with Σ notation. Here are some examples:

	Summation with Σ notation	No. of terms	Sequence
e.g.1	$\sum_{k=1}^{n} \frac{2k-1}{2^k}$	n	$\frac{1}{2}+\frac{3}{2^2}+\frac{5}{2^3}+...+\frac{2n-1}{2^n}$
e.g.2	$\sum_{k=1}^{n} k(2^{k-1})$	n	$1 + 2(2) + 3(2^2) + \ldots + n(2^{n-1})$
e.g.3	$\sum_{k=2}^{n} k(2^{k-1})$	$n - 1$	$2(2) + 3(2^2) + 4(2^3) + \ldots + n(2^{n-1})$
e.g.4	$\sum_{k=1}^{n+1} k(2^{k-1})$	$n + 1$	$2(2) + 3(2^2) + 4(2^3) + \ldots + (n+1)(2^n)$
e.g.5	$\sum_{k=1}^{n} (-1)^{k+1}k^2$	n	$3^2 - 4^2 + 5^2 - 6^2 + \ldots + (-1)^{n+1}n^2$
e.g.6	$\sum_{k=1}^{n+1} (-1)^{k+1}k^2$	$n + 1$	$3^2 - 4^2 + 5^2 - 6^2 + \ldots + (-1)^{n}(n+1)^2$
e.g.7	$\sum_{k=50}^{n} (-1)^{k+1}k^2$	$n - 49$	$-50^2 + 51^2 - 52^2 + 53^2 + \ldots + (-1)^{n+1}n^2$

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Mathematical Induction

Write out what is meant by the summation with Σ notation below.

	Summation with Σ notation	No. of terms	Summation with Σ notation
Q.1	$\sum_{k=1}^{n} \frac{1}{k(k+1)}$		
Q.2	$\sum_{k=1}^{n+1} \frac{1}{k(k+1)}$		
Q.3	$\sum_{k=10}^{n} \frac{1}{k(k+1)}$		
Q.4	$\sum_{k=n}^{2n} \frac{1}{k(k+1)}$		
Q.5	$\sum_{k=n+1}^{2(n+1)} \frac{1}{k(k+1)}$		

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HKDSE Questions (level 1)

1. ***[M2 2016 Q.5(b)]***

Given that $\sum_{k=1}^{n}(-1)^k k^2 = \frac{(-1)^n n(n+1)}{2}$ for all positive integers *n*.

Evaluate $\sum_{k=3}^{333}(-1)^{k+1}k^2$. ***(2 marks)***

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HKDSE Questions (level 1)

2. ***[M2 2018 Q.6(b)]***

Given that $\sum_{k=1}^{n} k(k+4) = \frac{n(n+1)(2n+13)}{6}$ for all positive integers n.

Evaluate $\sum_{k=333}^{555} (\frac{k}{112})(\frac{k+4}{223})$. ***(2 marks)***

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HKDSE Questions (level 2)

3. ***[M2 2020 Q.5]***

Given that $\sum_{k=1}^{n} \frac{1}{k(k+1)(k+2)} = \frac{n(n+3)}{4(n+1)(n+2)}$ for all positive integers n.

Evaluate $\sum_{k=4}^{123} \frac{50}{k(k+1)(k+2)}$. ***(2 marks)***

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HKDSE Questions (level 1)

4. ***[M2 2019 Q.5]***

Given that $\sum_{k=n}^{2n} \frac{1}{k(k+1)} = \frac{n+1}{n(2n+1)}$ for all positive integers n.

Evaluate $\sum_{k=50}^{200} \frac{1}{k(k+1)}$. ***(2 marks)***

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